

ASSUMPTIONS & DECISIONS LOG

Assumptions and Major Technical Decisions

Project Overview

This document records the key assumptions, engineering decisions, and analytical trade-offs made during the execution of the Singapore Airbnb Market Intelligence Analysis project. Documenting these decisions improves transparency, reproducibility, and stakeholder understanding while providing context for the analytical results presented throughout the project.

Assumption 1: Review Frequency as a Demand Indicator

One of the primary challenges encountered during the analysis was the absence of direct booking information. Since the dataset did not contain completed booking counts or transaction records, an alternative demand indicator was required.

To address this limitation, review frequency was used as a proxy measure of booking activity. Listings receiving reviews more frequently were assumed to experience higher levels of guest activity than listings receiving fewer reviews.

While review frequency does not perfectly represent actual booking volumes, it provides a reasonable approximation that enables demand-side analysis when direct booking data is unavailable.

Assumption 2: Revenue Estimation

Actual host earnings were not available within the source datasets. Therefore, an estimated revenue metric was created using listing prices and occupancy-related indicators.

The purpose of this calculation was not to determine exact financial performance but rather to support comparative analysis across listings and host categories.

Estimated revenue should therefore be interpreted as a relative performance indicator rather than a precise measure of host income.

Assumption 3: Missing Review Information

Several review-related attributes contained missing values. Rather than treating these observations as data quality issues, they were interpreted as meaningful business information.

Listings without reviews were assumed to represent either newly created listings or properties that had not yet accumulated sufficient guest activity.

This assumption helped preserve the integrity of the dataset while avoiding unnecessary data manipulation.

Assumption 4: Host Classification Strategy

To support host behaviour analysis, hosts were segmented according to the number of listings under management.

The following classification structure was adopted:

- Single Listing Host
- Small Portfolio Host (2–5 Listings)
- Medium Portfolio Host (6–20 Listings)
- Large Portfolio Host (20+ Listings)

This classification enabled meaningful comparison of marketplace participation patterns and supply-side dynamics.

Assumption 5: Professional Host Definition

Hosts managing more than five listings were classified as professional operators.

This decision was based on the assumption that hosts operating larger portfolios are more likely to treat Airbnb as a business activity rather than a casual sharing-economy participation model.

The classification supported analysis of pricing strategies and market concentration.

Key Engineering Decisions

Several engineering decisions were made throughout the project:

- Creation of a consolidated Listing Master Dataset
- Use of dimensional modeling principles
- Development of engineered business metrics
- Adoption of reproducible notebook-based workflows
- Use of visualization-driven exploratory analysis

These decisions prioritized analytical clarity, reproducibility, and business relevance.

Conclusion

The assumptions and decisions documented in this report were necessary to address data limitations while maintaining analytical integrity. Although alternative approaches could have been adopted, the selected methodology provided a practical balance between technical rigor and business value.